

Practical 5

Practice Lab Assignment

CODETANTRA

Home Learn Anywhere

202501040147@mitaoe.ac.in Support Logout

5.1.1. Stacked Plot

04:02

Stacked Plot

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

stackedpl...

Submit

Debugger

1 import matplotlib.pyplot as plt

2 import pandas as pd

3

4 # Data for Months and Temperature for three cities

5 data = {

6 'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July', 'August', 'September', 'October', 'November', 'December'],

Average time 0.220 s 220.00 ms

Maximum time 0.220 s 220.00 ms

1 out of 1 shown test case(s) passed

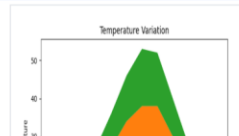
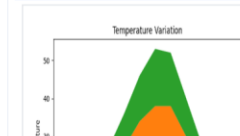
Test case 1 220 ms Debug

Expected output

Actual output

Terminal Test cases

Prev Reset Submit Next



Lab Assignment

CODETANTRA

Home Learn Anywhere

202501040147@mitaoe.ac.in Support Logout

5.2.1. Titanic Dataset

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:
The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the pandas library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:
To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:
Write the code to create a series of visualizations as follows:
Bar Plot (Pclass Distribution):

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset from the CSV file
5 df = pd.read_csv('titanic.csv')
6
7 # Set up the figure for 5 subplots
8 fig, axes = plt.subplots(3, 2, figsize=(12, 12))
```

Average time 0.968 s 968.00 ms Maximum time 0.968 s 968.00 ms 1 out of 1 shown test case(s) passed

Test case 1 968 ms Debug

Expected output Actual output

Terminal Test cases

CODETANTRA

Home Learn Anywhere

202501040147@mitaoe.ac.in Support Logout

5.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

- Use 30 bins for the histogram.
- Set the edge color of the bars to black (k).
- Label the x-axis as 'Age' and the y-axis as 'Frequency'.
- Add the title "Age Distribution" to the histogram.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	1
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.28		

Sample Data:
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.28

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
```

Average time 0.279 s 279.00 ms Maximum time 0.279 s 279.00 ms 1 out of 1 shown test case(s) passed

Test case 1 279 ms Debug

Expected output Actual output

Terminal Test cases

CODETANTRA

Home Learn Anywhere

202501040147@mitaoe.ac.in Support Logout

5.2.3. Bar plot of survival rate of passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

- Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).
- Set the chart type to 'bar'.
- Add the title "Survival Count" to the chart.
- Label the x-axis as 'Survived' and the y-axis as 'Count'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	1
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.28		

Sample Data:
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
```

Average time 0.308 s 308.00 ms Maximum time 0.308 s 308.00 ms 1 out of 1 shown test case(s) passed

Test case 1 308 ms Debug

Expected output Actual output

Terminal Test cases

CODETANTRA

HomeLearn Anywhere

202501040147@mitaoe.ac.inSupportLogout

5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

- Group the data by the **'Sex'** column, then use the **value_counts()** function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
- Use a **stacked bar chart** to display the survival counts.
- Add the title **"Survival by Gender"** to the chart.
- Label the x-axis as **'Gender'** and the y-axis as **'Count'**.
- The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
```

Average time0.395 sMaximum time0.395 s1 out of 1 shown test case(s) passed

Test case 1395 msDebug

Expected outputActual output

Survival by GenderSurvival by Gender

TerminalTest cases

CODETANTRA

HomeLearn Anywhere

202501040147@mitaoe.ac.inSupportLogout

5.2.5. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

- Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_counts()**.
- Use a **stacked bar chart** to display the survival counts.
- Add the title **"Survival by Pclass"** to the chart.
- Label the x-axis as **'Pclass'** and the y-axis as **'Count'**.
- The legend should indicate **'Not Survived'** and **'Survived'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
```

Average time0.353 sMaximum time0.353 s1 out of 1 shown test case(s) passed

Test case 1353 msDebug

Expected outputActual output

Survival by PclassSurvival by Pclass

TerminalTest cases

CODETANTRA

HomeLearn Anywhere

202501040147@mitaoe.ac.inSupportLogout

5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset. The chart should display the following specifications:

- Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
- Set the chart type to 'bar' and make it stacked.
- Add the title **"Survival by Embarked"** to the chart.
- Label the x-axis as **'Embarked'** and the y-axis as **'Count'**.
- Include a legend to distinguish between survivors and non-survivors (label the legend as **'Survived'** and **'Not Survived'**).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

BarPlotOf...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
```

Average time0.362 sMaximum time0.362 s1 out of 1 shown test case(s) passed

Test case 1362 msDebug

Expected outputActual output

Survival by EmbarkedSurvival by Embarked

TerminalTest cases

CODETANTRAHomeLearn Anywhere202501040147@mitaoe.ac.inSupportLogout

5.2.7. Box plot for Age Distribution08:00

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

- Use the **Pclass** column to group the data for the boxplot.
- Set the title of the plot to **"Age by Pclass"**.
- Remove the default subtitle with **plt.suptitle("")**.
- Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, S
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.28
```

BoxPlotF...Submit

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
```

Average time0.400 s400.00 msMaximum time0.400 s400.00 ms1 out of 1 shown test case(s) passed

Test case 1400 msDebug

Expected outputActual output

TerminalTest cases

PrevResetSubmitNext

CODETANTRAHomeLearn Anywhere202501040147@mitaoe.ac.inSupportLogout

5.2.8. Box Plot for Age by Survived02:00

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

- Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
- Set the title of the plot to **"Age by Survival"**.
- Remove the default subtitle with **plt.suptitle("")**.
- Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, S
```

BoxPlotF...Submit

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
```

Average time0.348 s348.00 msMaximum time0.348 s348.00 ms1 out of 1 shown test case(s) passed

Test case 1348 msDebug

Expected outputActual output

TerminalTest cases

PrevResetSubmitNext

CODETANTRAHomeLearn Anywhere202501040147@mitaoe.ac.inSupportLogout

5.2.9. Box Plot for Fare by Pclass01:47

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

- Use the **Pclass** column to group the data for the boxplot.
- Set the title of the plot to **"Fare by Pclass"**.
- Remove the default subtitle with **plt.suptitle("")**.
- Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, S
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.28
```

BoxPlotF...Submit

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
```

Average time0.399 s399.00 msMaximum time0.399 s399.00 ms1 out of 1 shown test case(s) passed

Test case 1399 msDebug

Expected outputActual output

TerminalTest cases

PrevResetSubmitNext

CODETANTRAHomeLearn Anywhere202501040147@mitaoe.ac.inSupportLogout

5.2.10. Scatter Plot for Age vs. Fare02:30

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to "**Age vs. Fare**".
3. Label the x-axis as '**Age**' and the y-axis as '**Fare**'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, S
2, 1, 1, "Cumings, Mrs. John Bradley (Florence Briggs Thayer)", female, 38, 1, 0, PC 17599, 71.28
3, 1, 3, "Heikkinen, Miss. Laina", female, 26, 0, 0, STON/O2. 3101282, 7.925, S
```

AgeFareS...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
```

Average time0.261 s261.00 msMaximum time0.261 s261.00 ms1 out of 1 shown test case(s) passed

Test case 1261 msDebug

Expected outputActual output



TerminalTest cases

PrevResetSubmitNext

CODETANTRAHomeLearn Anywhere202501040147@mitaoe.ac.inSupportLogout

5.2.11. Scatter Plot for Age vs. Fare by Survived02:30

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to "**Age vs. Fare by Survival**".
4. Label the x-axis as '**Age**' and the y-axis as '**Fare**'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Braund, Mr. Owen Harris", male, 22, 1, 0, A/5 21171, 7.25, S
```

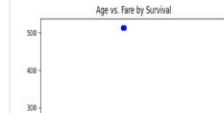
AgeFareS...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
```

Average time0.289 s289.00 msMaximum time0.289 s289.00 ms1 out of 1 shown test case(s) passed

Test case 1289 msDebug

Expected outputActual output



TerminalTest cases

PrevResetSubmitNext

